

Batch 1 (SecAB): Fundamentals + CPU + input/output devices

CONCEPT	FORMULA / RULE	TRAP
Data vs info + IPO	Data raw, info processed; IPO = keyboard to CPU to monitor.	Info-called-data; fast mistaken for smart.
Analog/digital/hybrid	Analog continuous, digital discrete, hybrid both.	Type mixed by example.
Micro to super	Micro/mini/mainframe/super by size-power; PARAM = Indian supercomputer.	Mainframe declared dead; sizes swapped.
Generations 1-5	G1 vacuum, G2 transistor, G3 IC, G4 VLSI/microprocessor, G5 AI.	G4-AI; tech confused with years.
Bits + units	4 bits = nibble, 8 = byte; KB/MB/GB x1024; A = 65 ASCII.	Byte-bit swap; 1000 vs 1024.
CPU/ALU/CU	ALU = math, CU = control; cycle fetch-decode-execute.	ALU/CU swapped; CPU called the box.
Registers/cache/board	Registers fastest (PC/MAR/MDR); cache SRAM levels; board holds CPU/RAM/slots.	RAM-cache swap; board = CPU.
Buses/clock/keys	Bus = data lanes; GHz + cores = speed; QWERTY layout.	Bus-library; cores-only speed.
Point/scan readers	Mouse points; scanner digitizes; OCR text, OMR bubbles, MICR cheques.	Scanner-printer swap; OCR-OMR mix.
Codes/biometric	Barcode lines, QR squares; biometric = finger/face/iris.	Barcode = QR; password-only.
Touch/monitor	Touch = input+output; LCD/LED display types.	Touch input-only; LCD/LED mixed.
Printers + plotter	Impact noisy (dot/daisy for forms); non-impact quiet ink/laser (photo); plotter draws vector maps.	Laser-impact; printer-plotter swap; speakers called input.

60-SECOND SPEED ROUND (cover answers, recall aloud)

Q: 3rd + 4th gen tech?	A: IC; VLSI/microprocessor
Q: Nibble + 1KB? A in ASCII?	A: 4 bits; 1024 bytes; 65
Q: Cheques + exam sheets + text?	A: MICR + OMR + OCR

Batch 2 (SecBC): Memory + storage + software + OS + Word basics

CONCEPT	FORMULA / RULE	TRAP
Hierarchy + RAM	Registers fastest down to disk; RAM volatile workspace for running programs.	More-RAM-called-storage; volatility missed.
ROM/PROM/cache	ROM firmware; PROM/EPROM/EEPROM programmable; cache L1/L2 SRAM.	ROM writable; levels missed.
HDD vs SSD	HDD spinning magnetic, cheap-big; SSD chip flash, fast shock-proof.	Faster-called-bigger; uses swapped.
Optical/flash	CD/DVD/Blu-ray pits of light; pen drive/card flash + OTG.	Magnetic-optical mix; pen called RAM.
Units + files	Bit/byte/KB/MB/GB/TB x1024; movie GB, song MB; MB vs Mb differ.	1000/1024 mix; MB = Mb.
System vs app	System = platform (Windows/Linux); app = tools (Word/browser).	Browser called OS.
Utility/driver	Utility helps (antivirus/backup); driver drives hardware; firmware lives on chip.	Driver-OS mix; firmware = software.
Open-source + OS jobs	Free code (Linux); OS manages process/memory/files.	Open = insecure; jobs missed.
GUI/files/boot	GUI click-icons vs CLI text; path + extension; POST then loader.	CLI-GUI mix; boot = login.
Word + extension	Word = full processor (.docx); Notepad = plain text.	.doc = .docx assumed.
Format + setup	Fonts/bold/align + A4/portrait/margins set before printing.	Defaults assumed; setup blind.
Find/replace/merge	Ctrl+H replace with preview; mail merge = fields + preview + print.	Replace-all blind; merged unpreviewed.

60-SECOND SPEED ROUND (cover answers, recall aloud)

Q: Fastest memory? Work memory?	A: Registers; RAM
Q: 1 GB in bytes?	A: 1024^3
Q: Platform + tools?	A: System; application

Batch 3 (SecCD): Excel + PowerPoint

CONCEPT	FORMULA / RULE	TRAP
Workbook/cells	Sheets + tabs; cell A1 address; ranges named.	Sheet confusion; range unnamed.
Relative vs absolute	Relative shifts on fill; \$A\$1 locked, \$A1/A\$1 half-locked.	\$-blind fill breaks.
Formula + errors	Starts with =; BODMAS order; #DIV/0 and #VALUE diagnosed.	No-=; calculator habit.
Core functions	SUM/AVERAGE/COUNT/MAX/MIN + args; IF logic with nest (=IF(A1>40,P,F)).	Manual add; args missed.
Sort vs filter	Sort orders (expand selection!); filter hides by criteria.	Half-sorted scramble; header mixed in.
Charts	Right chart + title + labels; pie for share, line for trend.	Pie-everything; 3D; untitled.
Excel shortcuts	Arrows navigate, F2 edit, F4 repeat/\$, F11 chart.	Mouse-only slowness.
PPT decks/layouts	.pptx; layouts + masters keep consistency.	Wall-of-text; layout ignored.
Transitions/show	Subtle moves; F5 from start, Shift-F5 current, Esc exit; rehearse.	Spinning effects; read-along; no rehearse.

60-SECOND SPEED ROUND (cover answers, recall aloud)

Q: Fill with \$A\$1 behaves?	A: Locked (no shift)
Q: Sort-half disaster fix?	A: Expand selection + header + reapply
Q: Slideshow from current?	A: Shift-F5; Esc exits

Batch 4 (SecCD): Internet + email + LAN/MAN/WAN

CONCEPT	FORMULA / RULE	TRAP
Internet vs WWW	Internet = network of cables/ISP; WWW = pages browsed.	Internet = Google.
Browser + search	Browser surfs (tabs/history); engine finds (quotes/site:/filetype:).	Address-search mix; top-result trusted.
URL + domain	Protocol/domain/path; TLDs .in/.com/.org/.edu/.gov; typo-squat watch.	Domain = site; TLD ignored.
HTTP/S + cookies	S + padlock = secure shopping; cookies sessions, cache speed; clear both.	HTTP-shop; cookies-called-virus.
Download + trio	Down vs up by direction; To/CC/BCC with BCC hiding; reply-all discipline.	Up-down swap; BCC = CC.
Attachments + spam	Size limits + virus scan; phishing = urgent + typos + hover-links.	Open-all; known-sender trust.
SMTP/POP3/IMAP	SMTP sends (25/587); POP3 downloads, IMAP mirrors/syncs.	Send-receive swap.
Etiquette + safety	Subject/greeting/brevity, no ALL-CAPS; strong + 2FA + backup.	Urgent-all; weak; no backup.
Network + LAN	Network connects/shares; LAN local cable/Wi-Fi.	LAN = Wi-Fi; internet-network mix.
MAN + WAN	MAN city fiber/campus; WAN world via ISP (internet itself).	Scope mixed; internet = WAN-only.

60-SECOND SPEED ROUND (cover answers, recall aloud)

Q: Net vs web?	A: Network; pages
Q: Shop safely + crumbs?	A: HTTPS; clear cookies/cache
Q: Send vs receive vs sync?	A: SMTP; POP3/IMAP

Batch 5 (SecDEF): Devices + protocols + malware + hygiene + DBMS

CONCEPT	FORMULA / RULE	TRAP
Topology + modem	Star-hub, bus-line layouts; modem converts digital-analog-digital via ISP.	Modem = router.
Router/switch/hub	Smart routes, smarter learns, dumb broadcasts.	Hub = switch.
IP/MAC/Wi-Fi	IP logical (v4 dots, v6 colons); MAC burned-in physical; Wi-Fi wireless.	IP-MAC swap; Wi-Fi = internet.
HTTP/FTP + suite	HTTP web, HTTPS lock, FTP file transfer; TCP/IP suite + DNS names.	S = speed; TCP = IP.
Mail protocols	SMTP/POP3/IMAP + ports; DNS resolves names.	Port guesses.
Virus/worm/trojan	Virus needs host file; worm self-copies network; trojan disguised gift.	All-called-virus; trojan self-copy.
Ransomware/phish	Locks for ransom (backup beats it); phishing baits links (hover + report).	Pay-fast; link-safe assumed.
Firewall/antivirus	Wall blocks by rules (on!); antivirus scans + updates.	Firewall off; expired signatures.
Password + 2FA	Long unique + manager; OTP/app/biometric second factor.	1234 + reuse + no 2FA.
Browse/patch/backup	HTTPS + padlock; update now; 3-2-1 backup copy + restore.	Later; public-Wi-Fi banking; no backup.
DBMS basics	Tables + queries beat Excel; MySQL/Oracle multi-user.	Excel-called-database.

60-SECOND SPEED ROUND (cover answers, recall aloud)

Q: Smart/smarter/dumb order?	A: Router, switch, hub
Q: Host + network + gift?	A: Virus; worm; trojan
Q: Lock + bait defense?	A: Backup; hover + report

Batch 6 (SecG): Database + cloud + AI/IoT + extensions + shortcuts I

CONCEPT	FORMULA / RULE	TRAP
Table/record/field	Rows = records, columns = fields; Students(ID,Name).	Field = record swap.
Key + SQL	Primary key unique + not-null (RollNo); SELECT with WHERE queries.	Duplicate key; SQL = Excel.
Cloud/SaaS	Rent-not-own on demand; Drive storage, Gmail SaaS apps.	Cloud = Wi-Fi; disk-only.
QR/UPI/e-gov	QR squares scan; UPI money; portals = services; fraud watch.	Barcode-QR mix; cash-only.
AI/ML/GenAI	AI narrow-smart; ML learns; GenAI creates from prompts.	AI thinks; prompt ignored.
IoT/robotics	Things + sensors (watch/home); scripts + arms automate.	Remote = IoT; robots alive.
Word/Excel ext	.docx/.doc docs; .xlsx/.xls sheets, .csv data.	Docx-doc mix; CSV = Excel.
PPT/PDF/text	.pptx decks; .pdf fixed; .txt plain.	PPT-PDF mix; PDF editable.
Media/pack/run	JPEG lossy vs PNG lossless; ZIP packs; EXE runs; CSV data.	JPG-PNG mix; zip = folder.
Copy/paste/cut/undo	Ctrl+C/V/X/Z; clipboard holds; Ctrl+Y redoes.	Mouse-only; delete-retype.
Save/print	Ctrl+S/P + autosave on; paper set before print.	Save-later loss.

60-SECOND SPEED ROUND (cover answers, recall aloud)

Q: Unique + query?	A: Primary key; SQL
Q: Sky + files + service?	A: Cloud; storage; SaaS
Q: Cut + undo + redo keys?	A: Ctrl+X/Z/Y

Batch 7 (SecG): Shortcut master revision

CONCEPT	FORMULA / RULE	TRAP
Find + select all	Ctrl+F finds, Ctrl+A selects all; never scroll-hunt or drag-select.	Manual find; slow selection.
Bold/italic/line	Ctrl+B/I/U formats instantly; toolbar confirms.	Menu-only formatting.
File ops	Ctrl+N new, O open, W close, H replace; dialog verified.	File-menu crawl.
Slideshow keys	F5 from start, Shift-F5 current, Esc exit + rehearse.	Menu-show fumble; no rehearse.
Drill groups	C/X/V/Z edit; S/P save-print; F/A find-all; B/I/U format.	Known-but-unpracticed slowness.

60-SECOND SPEED ROUND (cover answers, recall aloud)	
Q: Find + all keys?	A: Ctrl+F/A
Q: Bold + italic + underline?	A: Ctrl+B/I/U
Q: New + open + close + replace?	A: Ctrl+N/O/W/H
Q: Show keys?	A: F5 / Shift-F5 / Esc
Q: Exam-day rule?	A: Drill groups + mock speed